

FIRST DROP

The water that runs
before the shower begins

TEAM

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Problem framing only. No intervention is proposed in this deck.





THE MOMENT

Water is
flowing

Washing has
not begun

The interval we are investigating

Adults in UAE homes who shower with warm water and wait 3–5 minutes for it to arrive

Situation: the tap is on, cold water runs to the drain, and they wait

Outside the bathroom (“I’m waiting outside” — Interviewee 1)

or standing by (“I just let the water run... and just sit there” — Interviewee 2)

Context: UAE homes; interviewed face-to-face, 27 Sept 2026

01

Waiting for hot water to arrive

Water in the hot water pipes cools between uses.
Hot water must travel from where it is heated to the shower.

02

The user is getting ready

Water may keep running after
a usable temperature arrives.

ILLUSTRATIVE CALCULATION

30

seconds of flowing water

4.35

litres released

$$8.7 \text{ L/min} \times 0.5 \text{ min} = 4.35 \text{ L}$$

Interviews reported 3–5 minute waits, not 30 seconds.

$8.7 \text{ L/min} \times 4 \text{ min} = \text{about } 35 \text{ L per shower}$

Flow: RSB audit of 45 high-use villas, Abu Dhabi / Al Ain (2015), via ADDC/Tarsheed 2017.
30-second wait assumed. Local study rate, not a UAE average or measured shower loss.

2

recorded interviews
27 Sept 2026

5:15

minutes of recording
face-to-face, phone-recorded

SAME 5 QUESTIONS

Warm or cold?
How long do you wait?
How much is lost?
What do you do meanwhile?
What happens to the water?

WHO WE INTERVIEWED

Interviewee 1 — “Mr Ali”: older man · warm showers · waits 3–4 min · waits outside the bathroom · nothing collected

Interviewee 2 — “Ryan”: young man · warm showers · waits 3–5 min · stays in the bathroom · nothing collected

Interviewee 3: *recording (about 4 min) not received — nothing from it is used on these slides*

Four things we heard

“About 3 to 4 minutes.”

How long do you wait for warm water?
— Interviewee 1, older man · 00:22

“I just let the water run until it gets warm and just sit there.”

What do you do while you wait?
— Interviewee 2, young man · 00:40

“I’m waiting outside.”

What do you do while you wait?
— Interviewee 1, older man · 00:45

“No idea, but...
I would say at least a bucket.”

How much water is lost?
— Interviewee 1, older man · 00:34

01

Minutes, not seconds

We had assumed about 30 seconds.
They said 3–4 and 3–5 minutes: six to ten times longer.

02

Nobody measures the loss

“No idea... at least a bucket.” “I don’t know the exact number...”
The run-off is invisible while it happens.

03

Waiting looks different in each home

One leaves the room (“I’m waiting outside”); one stays (“just sit there”).
Either way, the water goes down the drain.

One shower's warm-up, scaled to a year

$8.7 \text{ L/min} \times \text{wait (min)} \times 1 \text{ shower/day} \times 365 \text{ days}$

9,500

litres per year

3 min wait

low end reported

26.1 L per shower

12,700

litres per year

4 min wait

midpoint of 3–5 min

34.8 L per shower

15,900

litres per year

5 min wait

high end reported

43.5 L per shower

For comparison, the 30-second wait we first assumed would be about **1,588 L per year**.

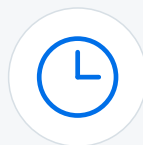
Adults in UAE homes who shower with warm water **need a way to** stop the 3–5 minute warm-up from sending clean water down the drain **because** the wait is passive time — they step out or wait it out — and nobody sees how much is lost.



Turn on the shower



Cold water to the drain



Wait 3–5 minutes



Shower

Why the water keeps running

01

HABIT

Turn it on, then wait — outside the room or standing by. The run-off is never handled.

Interviews 1–2

02

COST SIGNAL

Neither person could estimate the loss (“at least a bucket”). Nothing in the moment makes it visible.

Interviews 1–2 · context

03

SYSTEMS

Water is heated far from the shower; the pipe must clear every time. Changing the plumbing is rarely an option.

EPA 2026 · to confirm

04

CLIMATE

Colder inlet water in winter should lengthen the wait. Not covered in either interview.

To verify — interview 3

05

ROUTINE

The wait sits inside getting ready; extra steps compete with time. Not raised by either interviewee.

Assumption — to verify

Elsewhere (scheduled supply in Jordan, RV tanks) the same wait draws on a finite reserve — context, not a finding about our users.

None of today's options fits these users

What users do or could do

Why it doesn't work for these users

Turn it on and walk away

Interviewee 1: "I'm waiting outside."

Nobody is in the room when warm water arrives, so the run continues until they return.

Stand by and wait

Interviewee 2: "just sit there."

Passive minutes with the water in view – and it still goes down the drain.

Collect it in a bucket

Interviewee 1: "it just goes down the drain."

Neither interviewee does it. Extra handling and storage, and one user is not even in the room.

Thermostatic shut-off valve

Published approach (EPA 2026)

Stops the flow once warm water arrives; the cold slug still drains first, and it needs installing.

Recirculation loop

Published approach (EPA 2026)

Keeps the pipe warm, but it means re-plumbing the home (not an option for tenants; tenure not asked) and adds an energy cost.

We assumed a 30-second wait

...and a few litres lost: 4.35 L per shower.
A minor annoyance, not a routine.

Our starting assumption, before any interview (see slide 5).

“About 3 to 4 minutes.”

— Interviewee 1 · 00:22

“Three to five minutes.”

— Interviewee 2 · 00:16

“No idea... I would say at least a bucket.”

— Interviewee 1 · 00:34

“I’m waiting outside.” / “...and just sit there.”

— Interviewee 1 · 00:45 / Interviewee 2 · 00:40

6–10×

longer than we assumed
3–5 min reported vs 30 s

BEFORE

Residents waste water while waiting for hot water.

Any resident, anywhere
No estimate of the wait
No reason it persists
Nothing a team could act on

AFTER

Adults in UAE homes who shower with warm water **need a way to** stop the 3–5 minute warm-up from sending clean water down the drain **because** the wait is passive time — they step out or wait it out — and nobody sees how much is lost.

Specific user · reported wait (3–5 min)
self-described behaviour · a reason it persists

Questions?

OUR POINT OF VIEW

Adults in UAE homes who shower with warm water **need a way to** stop the 3–5 minute warm-up from sending clean water down the drain **because** the wait is passive time — they step out or wait it out — and nobody sees how much is lost.